

Guided Altium Exercise

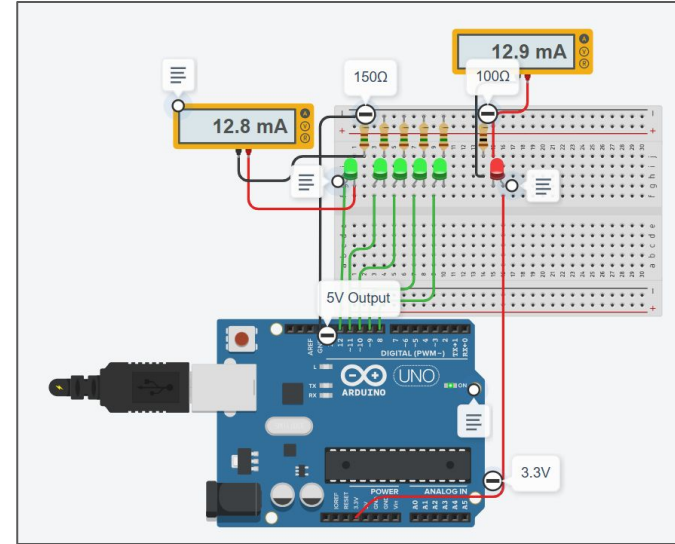
2 Boards; 5 LED Channels per Board

Practice PCB

A microcontroller can send **output signals** in most pins. For digital outputs, the **logic voltage** is either a HIGH or LOW. As voltage changes, resistance must change to meet the **operational current** for the LEDs. $(10mA < i < 20mA)$

<u>MCU-Devboard</u>	V_{HIGH}	V_{LOW}
Arduino-Uno	5V	0V
Stm32F401RE-Nucleo	3.3V	0V

[TinkerCAD](#) simulation



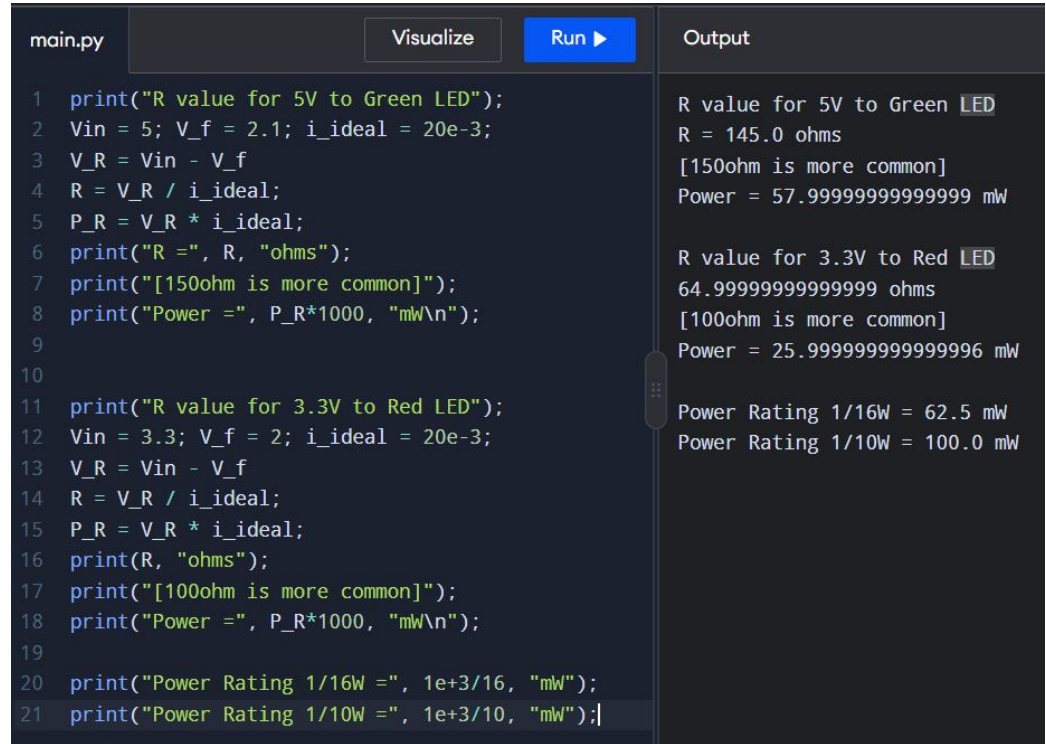
Let's create two boards that support both Arduino's and STM32's logic voltage outputs. These boards will also make the circuit significantly smaller with the absence of a breadboard

Part Selecting: Resistance and Power Rating

The Forward Voltage of [LTST_C190KRKT](#) and [LTST_C190GKT](#) are found on the datasheet

Use a [150ohm](#) resistor with a 1/10 power rating for the 5V circuit

Use a [100ohm](#) resistor with a 1/16 power rating for the 3.3V circuit



The image shows a Python IDE window with a file named 'main.py'. The code calculates the required resistor value and power rating for two different LED circuits. The first circuit is for a 5V supply and a Green LED (LTST_C190KRKT). The second circuit is for a 3.3V supply and a Red LED (LTST_C190GKT). The output shows the calculated resistor values and power ratings, along with common resistor values for reference.

```
main.py Visualize Run ▶  
1 print("R value for 5V to Green LED");  
2 Vin = 5; V_f = 2.1; i_ideal = 20e-3;  
3 V_R = Vin - V_f  
4 R = V_R / i_ideal;  
5 P_R = V_R * i_ideal;  
6 print("R =", R, "ohms");  
7 print("[150ohm is more common]");  
8 print("Power =", P_R*1000, "mW\n");  
9  
10  
11 print("R value for 3.3V to Red LED");  
12 Vin = 3.3; V_f = 2; i_ideal = 20e-3;  
13 V_R = Vin - V_f  
14 R = V_R / i_ideal;  
15 P_R = V_R * i_ideal;  
16 print(R, "ohms");  
17 print("[100ohm is more common]");  
18 print("Power =", P_R*1000, "mW\n");  
19  
20 print("Power Rating 1/16W =", 1e+3/16, "mW");  
21 print("Power Rating 1/10W =", 1e+3/10, "mW");
```

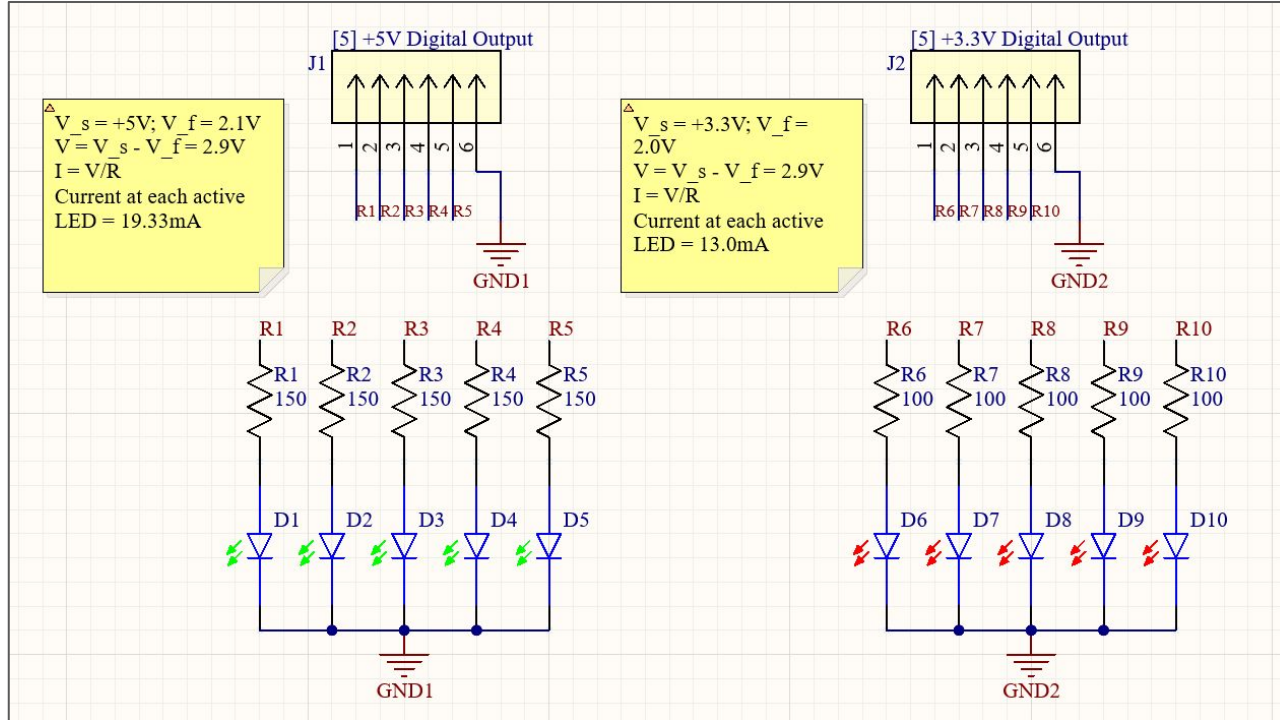
Output

R value for 5V to Green LED
R = 145.0 ohms
[150ohm is more common]
Power = 57.99999999999999 mW

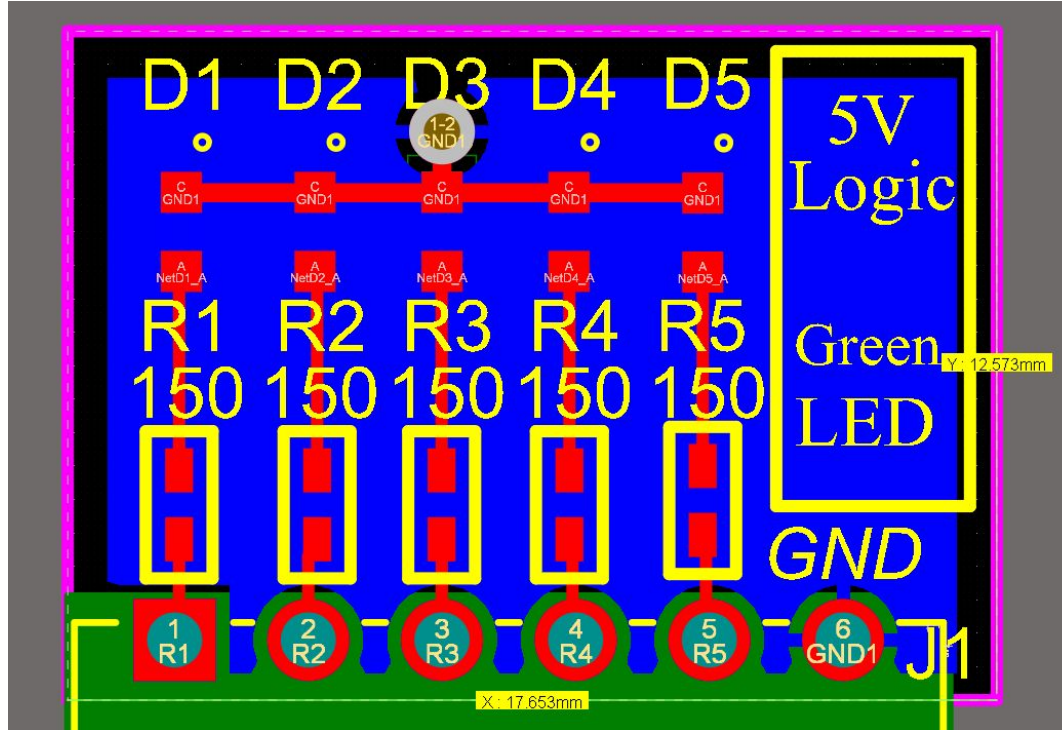
R value for 3.3V to Red LED
64.99999999999999 ohms
[100ohm is more common]
Power = 25.999999999999996 mW

Power Rating 1/16W = 62.5 mW
Power Rating 1/10W = 100.0 mW

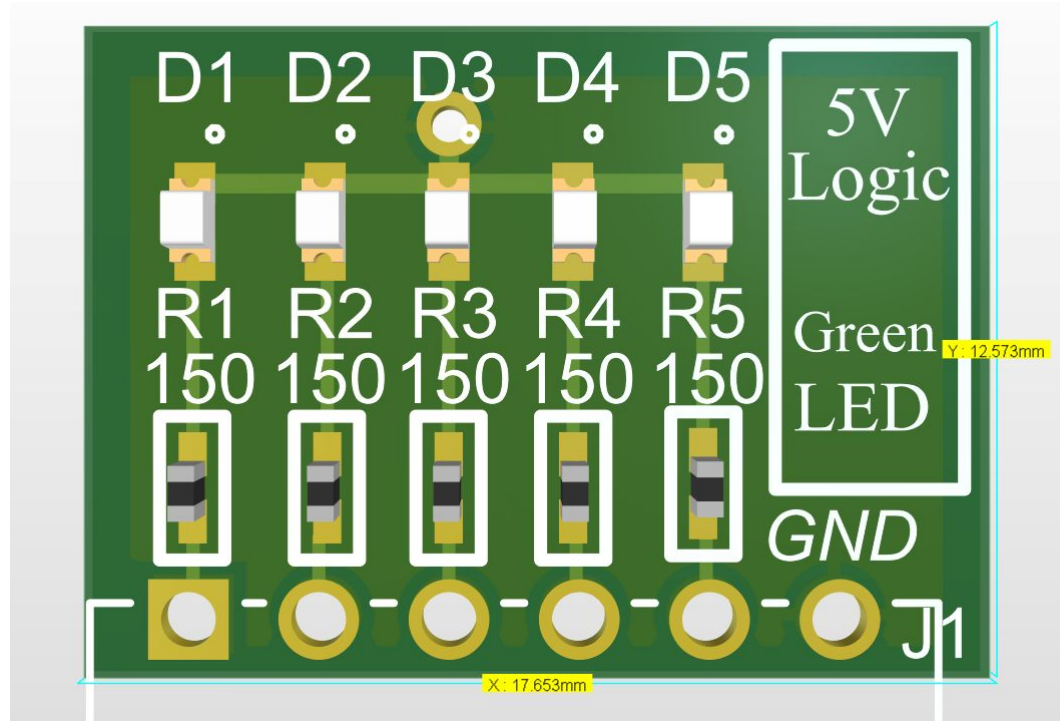
Schematic Sheet of Both Circuits



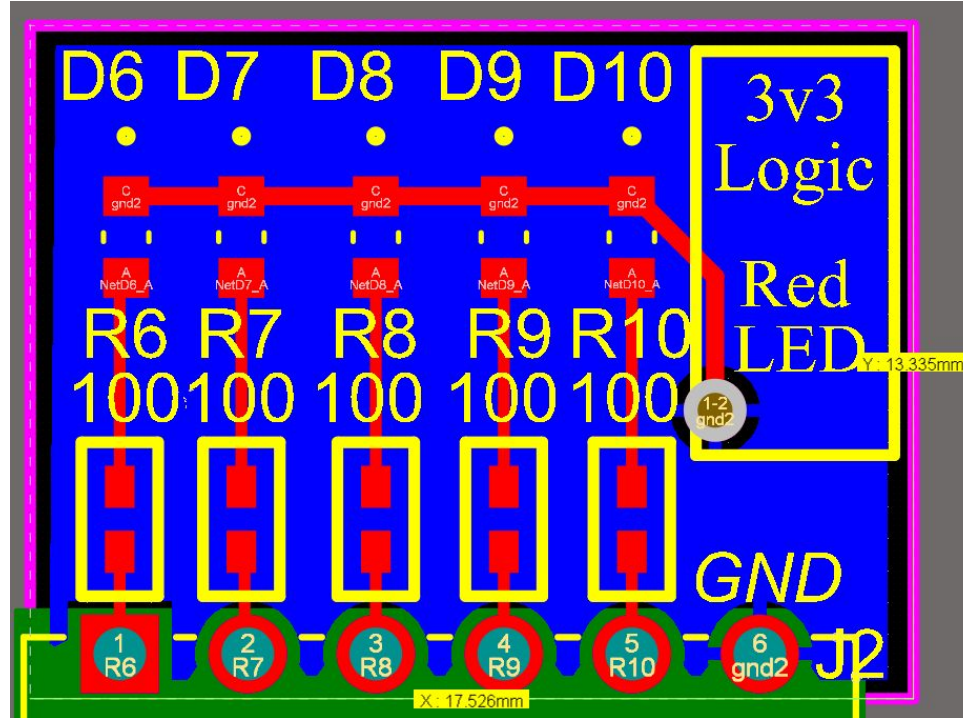
Board1 2D View



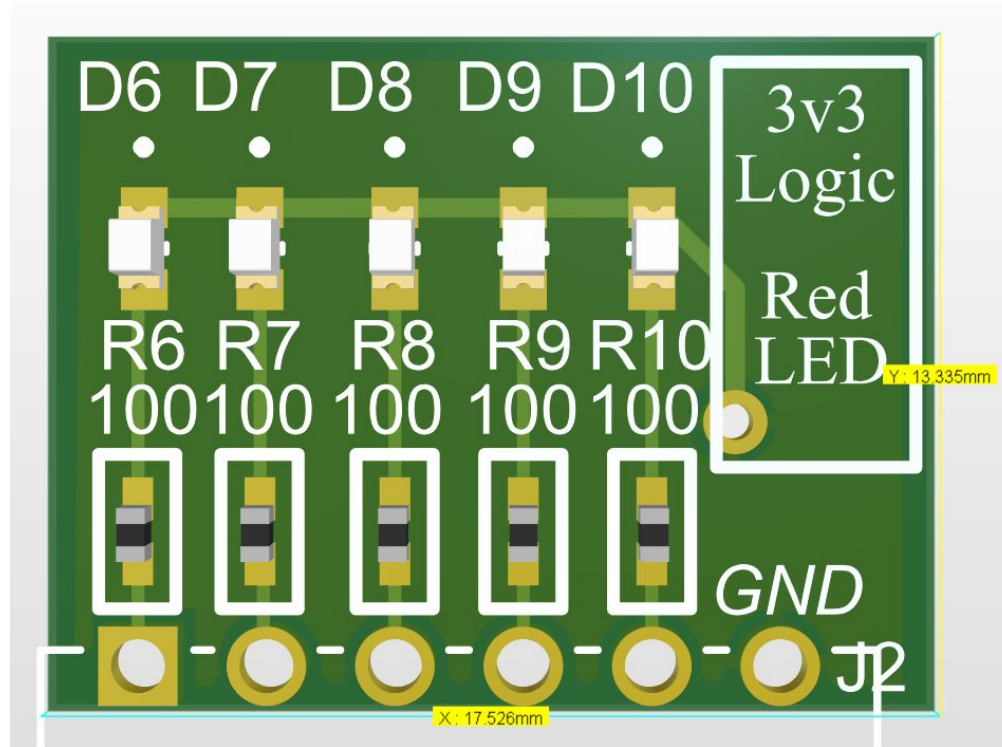
Board1 3D View



Board2 2D View



Board2 3D View



Practice BOM

<u>Part Number</u>	<u>Description</u>	<u>Designator</u>	<u>Qty</u>	<u>Availability</u>	<u>Manufacturer</u>	<u>Purchase Link</u>	<u>Cost per part</u>	<u>Part cost*Qty</u>	<u>Total Cost</u>
LTST_C190GKT	Green 569nm LED 2.1V 0603 (1608 Metric)	D1, D2, D3, D4, D5	5	In Stock	Lite-on	[link]	N/A	N/A	N/A
LTST-C190KRKT	Red 631nm LED 2V 0603 (1608 Metric)	D6, D7, D8, D9, D10	5	In Stock	Lite-on	[link]	N/A	N/A	
PPTC061LGBN-RC	6 Pos Header Connector (2.54mm) Through Hole, Right Angle	J1, J2	2	In Stock	Sullins	[link]	N/A	N/A	
ERJ-2RKF1500X	150 Ohms ±1% 0.1W, 1/10W Chip Resistor 0402 Thick Film	R1, R2, R3, R4, R5	5	In Stock	Panasonic	[link]	N/A	N/A	
CRCW0402100RFKE D	100 Ohms ±1% 0.063W, 1/16W Chip Resistor 0402 Thick Film	R6, R7, R8, R9, R10	5	In Stock	Vishay	[link]	N/A	N/A	